| Personal Homepage || ydz627@gmail.com || +1 (408)564-1069 |
| LinkedIn: @halffrost || Github: @halfrost |
Github World Star Ranking: 214, World Followers Ranking: 120
Professional Membership of ACM / IEEE / IEEE-CS / CCF / Sigma Xi

Dezhi Yu

EDUCATION

University of California, Berkeley — Berkeley, CA
Master of Information and Data Science, GPA: 3.966/4.00 — Jan. 2024 – Dec.2025

Stanford University — Palo Alto, CA
Graduate Certificate, Artificial Intelligence Specializations, GPA: 4.00/4.00 — Jan. 2023 – Dec.2023

Carnegie Mellon University — Pittsburgh, PA
Master of Science in Software Engineering, GPA: 3.6/4.00 — Sep. 2021 – Dec.2022

Huzhou University — Zhejiang, China
Bachelor of Science in Computer Science, GPA: 3.8/4.00, Ranking: 3/150 — Sep. 2009 – Jun. 2013

PUBLICATIONS

- **"Efficient Cross-GPU Communication for Disaggregated LLM Serving"**
- **"KVDirect: Distributed Disaggregated LLM Inference"** Arxiv
  Submitted to 2025 IEEE Transactions on Parallel and Distributed Systems (IEEE TPDS 2025).
- **"How Far Are Video Models from True Multimodal Reasoning?"**
  Accepted by ECCV 2026 The 19th European Conference on Computer Vision.
- **"ROSE: A Reward-Oriented Data Selection Framework for LLM Task-Specific Instruction Tuning"** Arxiv
  Accepted by Findings of 2025 Empirical Methods in Natural Language Processing (EMNLP 2025).
- **"LaySummX at BioLaySumm: Retrieval-Augmented Fine-Tuning for Biomedical Lay Summarization Using Abstracts and Retrieved Full-Text Context"** ACL workshop
  Accepted by the 24th Workshop on Biomedical Language Processing (BioNLP, ACL 2025).
- **"Direct Preference Optimization for Chatbot Fine-Tuning: An Empirical Study"** Arxiv
- **"Generative Modeling of Bach-Style Symbolic Music: A Comparative Study of Autoregressive, Latent-Variable, and Adversarial Approaches "** Arxiv

Academic Service

Program Committee & Reviewer

- Program Committee Member: ICLR 2026, AISTATS 2026, ACL ARR (2025–2026), AAAI, IJCAI, KDD Workshops
- Reviewer: NeurIPS (2024, 2025 Workshops, 2026), IEEE SSCI 2025, KSEM 2025, IEEE MIPR 2025

Journal Reviewer

- IEEE Transactions on Image Processing (TIP), IEEE Transactions on Circuits and Systems for Video Technology (TCSVT), Journal of Medical Imaging (JMI), Journal of Machine Learning for Biomedical Imaging, ACM Transactions on Internet Technology (TOIT)

TECHNICAL SKILLS

LLM systems: Disaggregated serving, inference runtime optimization, KV-cache/memory management, cross-GPU communication, RDMA, InfiniBand/RoCE/EFA, multi-cluster deployment, capacity planning, reliability.

Post-training and evaluation: SFT/DPO, reward-oriented data selection, instruction tuning, RAG, model evaluation pipelines, multimodal reasoning benchmarks.

Infrastructure: Kubernetes, Krew, Helm, Istio, Envoy, Prow, Bazel, AWS Cloud, Docker, Linux, gRPC, Apache Thrift, Protobuf, Beego, Kafka, Redis, MySQL, PostgreSQL, Hive, Spark, Blink, Phabricator, Git, Jenkins, GraphQL.

Languages: Go, Python, Java, C/C++, Rust, SQL, JavaScript, Shell.

PROFESSIONAL EXPERIENCE

Bytedance — **San Jose, CA**
Senior Software Engineer - Large Language Model — Feb 2023 – Now

- Led inference system engineering for **Volcengine Ark / MaaS 2.0**, a commercial Model-as-a-Service platform comparable to Amazon Bedrock and Google Vertex AI. Optimized large-scale LLM and multimodal model serving across distributed clusters, improving reliability, resource efficiency, and production scalability while supporting approximately **7T tokens/day** and **210T+ tokens/month**.

- Built **0→1 data pipeline infrastructure** for the company-wide AI feedback and Case Platform, ingesting logs from Ark EP and ArkClaw, processing events with Flink, and persisting structured cases into a Lance-based data lake; supported approximately **516M events/day** to power downstream LLM evaluation, analytics, and data flywheel systems. [link]
- Architected **PromptPilot**, a 0→1 context-engineering and LLM platform infrastructure for prompt optimization, model benchmarking, and AI application experimentation. Designed control plane / data plane separation, traffic routing, resource governance, secure multi-tenant isolation, traffic segmentation, and compute scheduling layers; adopted by **983 teams cumulatively**, supporting **25K+ optimization tasks biweekly**, **6K+ model evaluation tasks biweekly**, and **5.62T tokens consumed over the past 6 months**. [link]
- Designed and productionized **multimodal harness infrastructure** for VideoPilot, converting Wan 2.6 and SAM Audio Large models into scalable vefaas-based serving endpoints. Built reusable orchestration and evaluation workflows for model invocation, generation-result collection, quality assessment, and content ranking, supporting **2K+ generation tasks/day**, reducing manual evaluation effort by **67.8%**, enabling scalable iteration across **16+ multimodal models or pipelines**, and contributing to **￥20M+ annual revenue**. [link]

Ant Group — **Mountain View, CA**

Summer Of Code Mentee, Software Engineer Intern — May 2022 – Oct. 2022

- Designed and implemented multiple vm's **Envoy WASM** extension based on **cgo** and **tcp filter**, which supports traffic governance over **HTTP** traffic and **gRPC** services, and configured rules by **Kubernetes CRD** standard; designed and implemented a high-performance **message queue** that can synchronize messaging between multiple wasm **virtual machines**.
- Used kprobe, uprobe, USDT to mointor **eBPF** program that can trace the TCP protocol stack and added NetFilter to the network of processes in the same cgroup, which filters 70 million requests per day and improves filtration efficiency by 6 times.
- Developed the **eXpress Data Path** program, which is based on **LSM** feature of Linux 5.7, to discard data packets before it enters the protocol stack, thus reducing system resource consumption by illegal network requests, and strengthening container security.
- Designed and implemented a set of **eBPF** programs in **kernel mode** and **user mode**, based on co-re feature on Linux Kernel 4.9+, with the BCC + libbpf toolkit; took the data stored in **Maps** in the **Linux kernel mode**, analyzed BPF map in the user space program, and significantly reduced the cost of memory by 29.7%.

The Linux Foundation — **Mountain View, CA**

LFX Mentee, Software Engineer Intern — May 2022 – Oct. 2022

- Designed and implemented **Kubernetes-**native APIs of **Karmada** with high availability and extensibility in provision of advanced Cluster Resource Modeling capabilities to offer better scheduling in **multi-cloud** scenarios.
- Collaborated with 3 teams and proactively led 7 people to discuss, verify and fix users' pain points in production environment.
- Optimized algorithms to improve the scheduling performance and scheduling ability from **300** resourcebindings to **10k** resourcebindings per second for **60k** of clusters and **3000k** of nodes.

Binance.com — **Singapore, Singapore**

Staff Engineer — May 2021 – Oct. 2021

- Led a team consisting of 7 engineers to implement from scratch and successfully launch the firm's first Strategy Distribution Engine – **Themis**, a smart strategy engine based on traffic flow and predicate conditions' strategy.
- Designed architecture of service mesh cloud native app on top of **Golang** backend services, using **MySQL** and **Redis** as database, **Prow**, **Bazel** as CI/CD, **Hive**, **ClickHouse** as data statistics and **AWS**, K8s + Istio as deployment environment.
- Led the architecture redesign of the **A/B testing platform**, improving test setup efficiency by 40%. This involved developing a new modular system that enables seamless integration with multiple data sources and **machine learning models**, supporting over 500 concurrent experiments.
- Enhanced the platform's scalability to handle a **300% increase** in experimentation volume, supporting a peak load of 2,000 experiments per month without compromising on performance. Achievements include optimizing the data processing pipeline and implementing dynamic resource allocation algorithms.
- Spearheaded the integration of machine learning algorithms(**Random Forests/K-Means/Neural Networks**) to automate the segmentation and targeting process for experiments. This initiative improved the accuracy of user targeting by 25% and increased the overall impact of experiments on user engagement metrics by 15%.
- Developed advanced **data analysis and visualization tools** that reduced the time to insight from experimentation data by 50%. These tools provide real-time access to key metrics and performance indicators, enabling product teams to make informed decisions rapidly.
- Themis ecosystem managed to accumulate **200 million** users within **10 days** of its release to the public, with delivery rate above **97%,** daily peak value over **5000k** and online connections averaged **10k-15k** QPS.

Alibaba Cloud — **Shanghai, China**

Senior Backend Engineer | Team Leader — Feb 2017 – Jun 2020

- Participated in Tmall Double Eleven online shopping promotion for 2 consecutive years; led team consisting of 5 engineers.

- Enriched **Taco ecosystem** by implementing **Taco V2**, an infura-like API gateway on top of Golang backend services, which used **MySQL** and **Redis** as high-performance database, **RabbitMQ** and **Kafka** as message queue, **Hive**, **Blink** and **Elasticsearch** as data and message pipeline query, and **gRPC**, **Apache Thrift** and **HTTP** as communication protocol.
- Taco ecosystem successfully handles **1100 million+** push notifications per day, with delivery rate above **97%** and delivery time of **0.72-1.2 seconds**; daily peak value can reach **750k** online connections with **30k-50k QPS**.

Function Finance **Shanghai, China**
Infrastructure Team Lead May 2016 – Feb 2017

- Led engineer team to designed and implement a **FaaS infrastructure** snapshot system using Java. Speed up the average response time by 65% and improved the service robustness by 81% via parallelization and **local caching** techniques for read-heavy operations with PB level daily caching volume.
- Migrate 12 backend docker service to AliCloud, a cloud **kubernetes platform** and refactored the infrastructure by using redis and **mysql** in AliCloud, which increased the stability of the company's entire back-end infrastructure by 65%.
- Designed and implemented a internal system to intelligently match projects for investors, utilizing a **recommendation system model**, text abstraction model, and Text-to-Speech model, increasing the success rate of project bidding by 40%.

PINGAN Tech **Shanghai, China**
Infrastructure Team Lead Nov 2015 – Apr 2016

- Designed and implemented the 1QB Wallet recommendation system, matching clients and products based on the keywords of previously searched items to boost sales, using **RPC** and **HTTP** framework in **Java**, improving the accuracy by 30%.
- Building new Code Review platform from scratch; developed Java package deployment and remote backup distribution platform, including **Jenkins** integrated static code scanning on daily basis and **CI/CD pipeline**, increasing efficiency by 5 times.
- Developed a trading system in Go, leveraging the **Kafka** Messaging System, to enable users' participation in flash deal ordering events, enabling millions of users to engage in **millisecond-level** promotional activities. In the first year since the release and launch of this system, it has contributed an incremental revenue of **10 Billion RMB** in GMV to the company.
- Refactored the task queue and task manager using Celery, **RabbitMQ** and **MySQL** to support larger-scale, distributed tasks; Queue latency reduced by 20%, Consume rate increased by 4x and Memory used decreased by 40%.

Quatanium Tech **Shanghai, China**
Research Engineer Oct 2013 – Nov 2015

- Designed and implemented firm's first Cluster Observability Platform from scratch. Built a log service based on logstash, reformat, and transfer logs from grouping system to Kibana log platform, handing million logs per day with very high stability and availability which annual availability rate reaches 99.999%; Created Grafana dashboards to report the counters of logs.
- Designed and implemented CI/CD pipelines with YAML and deployed Docker containers to AWS, making them automatically build, pack, and deploy the log service to production, serving an average of 200 integrations per day.
- Designed and implemented an alert service for error logs, which can report the errors and create fix tickets for DevOps engineers automatically, reducing fix waiting time by 80%.

SELECTED PERSONAL OPEN-SOURCE PROJECTS

- **LeetCode-Go (33.8K star)**: A book on LeetCode solutions in Go, featuring 100 percentile test coverage and 100 percentile runtime performance. Tthe book, has 6.99 million PV, 3.83 million UV, and 174K PDF downloads. [Github Link]
- **Halfrost-Field (13.3K star)**: A personal technology blog that has cumulative 8.14 million PV and 4.54 million UV. [Github Link]
- **vue-objccn** (2.0K star): Used Vue.js to develop cross-platform full stack application, 2.7K downloads. [Github Link]
- threes-ai: Deep Reinforcement Learning for the Threes! game. [Github Link]